DO NOW — RECALL

Floating-point subtraction

Name: _____ Class: _____ Date: _____

1. Why must floating-point exponents match before the mantissas are added directly?

2. Convert 01100000 1110 to denary. Use an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

3. Convert $6D_{16}$ to eight-bit binary.

4. Use 8-bit two’s complement to calculate 18 − 29 . Show how subtraction becomes addition.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Because corresponding mantissa bit positions must represent the same scaled place values . Aligning the exponents puts both operands on the same scale before the mantissas are added.

2. Model answer

0.1875 Mantissa 01100000 = 0.75 ; exponent 1110 = −2 ; $0.75 \times 2^{-2} = 0.75 \div 4 = 0.1875$.

3. Model answer

01101101_2 6 → 0110 and D → 1101 .

4. Model answer

+18 = 00010010 +29 = 00011101 −29 = 11100011 00010010 + 11100011 = 11110101 11110101 represents −11 . So 18 − 29 = 18 + (−29) = −11 .

By the end, you can…

• align exponents and subtract by adding a negated mantissa;

• handle positive, negative and zero differences;

• retain working precision and normalise after cancellation.

Vocabulary — know these terms

Recall the term — Negation

Recall the term — Cancellation

Recall the term — Guard bits

Recall the term — Arithmetic right shift

Recall the term — Renormalisation

Recall the term — Underflow

NEW LEARNING

Read, annotate and explain

A − B = A + (−B)

1. Align exponents
2. Keep guard bits
3. Negate B's mantissa
4. Add
5. Normalise

Negate a two’s-complement mantissa by inverting the working bits and adding 1.

Do not negate the exponent. It controls scale, not the sign of the operand.

Align before subtracting

Choose a common exponent and use arithmetic right shifts to align the mantissas. Keep any low-order guard bits needed for an exact working result.

Negate the second mantissa

At the chosen working width, invert every bit of B’s aligned mantissa and add 1.

Add the aligned values

Sign-extend both operands to the same working width, add them, and interpret the signed result using the common exponent.

Handle cancellation, zero and range

A small difference may need left shifts and a reduced exponent to become normalised. Equal operands produce zero.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Subtract 1.5 from 5

Use common exponent 3

Align both mantissas using sign-preserving right shifts.

```
A: 0010100
B: 0000110
```

Invert and add 1

```
B:        0000110
invert →  1111001
add 1  →  1111010
```

Negate the aligned mantissa, not the exponent.

Calculate A + (−B)

```
0010100
 + 1111010
 = 0001110
```

The working difference is $14/32 \times 2^3$.

Shift left and reduce the exponent

```
0001110 × 2³
→ 011100 × 2²

Final: 011100 0010
```

Check: $5 - 1.5 = 3.5$.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Build the answer

Guided 2 — CHECK

Build the answer

Guided 3 — CHECK

Build the answer

Guided 4 — CHECK

Build the answer
Recognise the all-zero result and stop; it does not need further normalisation.

Guided 5 — CHECK

Build the answer

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

When subtracting B from A, only B's mantissa is negated before addition.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY: 2. Negative difference: 1.5 − 5

Use a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Give the final answer in normalised floating-point form before checking.

Method / working:

Guided 2 — TRY: 3. Subtract a negative: 5 − (−1.5)

Use a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Give the final answer in normalised floating-point form before checking.

Method / working:

Guided 3 — TRY: 4. Cancellation: 6.5 − 6

Use a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Give the final answer in normalised floating-point form before checking.

Method / working:

Guided 4 — TRY: 5. Equal operands

Use a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Give the final answer in normalised floating-point form before checking.

Method / working:

Guided 5 — TRY: OCR JUNE 2017 · Q6(d) · [6]

Use a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Give the final answer in normalised floating-point form before checking.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Question 1 · ADAPTED OCR EXAM PRACTICE · [6]
Calculate 010010 0010 minus 010010 0100 . Show exact alignment, negation, addition and an answer in normalised floating-point form using a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent.
Adapted from OCR June 2017 Q6(d): operand order reversed; classroom marks.

Question 2

Question 2 · ADAPTED EXAM PRACTICE · [5]
Calculate 010010 0100 minus 011010 0011 . Use a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Show alignment and negation, then give the answer in normalised floating-point form.
Adapted from the OCR subtraction task: changed values; classroom marks.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Question 3 · ORIGINAL EXAM-STYLE EXTENSION · [5]
Calculate 011000 0010 minus 100000 0001 using a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Show how you handle the negative second operand and give the final answer in normalised floating-point form.

Question 4

Question 4 · ORIGINAL EXAM-STYLE EXPLANATIONS · [4]
State the stored result when a value is subtracted from itself, using a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. [1]
Explain why normalising a small difference can require a lower exponent. [2]
Explain why a nonzero low-order bit must be retained during alignment where possible. [1]